MATHEMATICAL APPENDIX TO HV27 v08 · PREPRINT

Counting five-label records and detecting one-step errors on the Sun–Moon torus

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HV27 counted 6,264 faces for an equal grid with a common origin [1, §6]. We show that its faces give 6,264 distinct records of five angular labels. A uniformly chosen one-step edit produces an impossible record with probability 1223/1740. Two fixed lunar-mansion partitions and an equal grid shifted by +0.1° each give 6,597 distinct records and probability 508/733.

1. The five labels and the edit rule

Measure angles in turns. Let the sidereal longitudes of the Moon x and Sun y vary independently on the circle. Write $h = x - y$ and $s = x + y = 2x - h$, modulo one. Every bin includes its left endpoint and excludes its right endpoint in increasing cyclic order. The record has five coordinates:

- **Nakshatra N.** Partition x into 27 consecutively numbered arcs with distinct names, each of positive width less than 1/12 turn.
- **Lunar rashi R.** Partition x into 12 equal bins beginning at zero.
- **Yoga Y.** Partition s into 27 equal bins beginning at zero.
- **Karana K.** Set $j = \lfloor 60h \rfloor$, taking $0 \le h < 1$. At $j = 0$ the label is 0. At $j = 1,\ldots,56$ it is $1 + ((j - 1) \bmod 7)$. At $j = 57,58,59$ it is 8,9,10 respectively.
- **Tithi T.** Take $\lfloor j/2 \rfloor \bmod 15$. This coding records the ordinal within a fortnight and suppresses the fortnight.

The traditional five limbs include the weekday [9]. Here the record uses lunar rashi in its place. For K, an edit follows the cyclic index order 0,...,10 (Kimstughna, the seven movable names in their usual sequence, Shakuni, Chatushpada, Naga). This order specifies which changes count as errors of one step. For N, R, Y and T, use their natural cyclic order.

Select one of the C distinct realizable records uniformly. Choose one of its five coordinates uniformly, then increase or decrease its label by one with equal probability, wrapping at either end of the label order. Write D for the probability that the edited record is unrealizable. Every realizable record has equal weight, irrespective of its angular area or its frequency in observations.

Let v count distinct boundaries in the union of the N and R partitions. Let q count the members of this union on the lattice of integer multiples of 1/1080 turn (one third of a degree).

Theorem. The number of distinct realizable records equals the number of open faces and is C = 3240 + 87v − 3q. Under the stated edit rule, the four fixed partitions give the following values of D.

N partition	v	q	C	D
Equal arcs, origin zero	36	36	6,264	1223/1740
H20 fixed J2000 boundaries	39	12	6,597	508/733
H21 fixed J2000 boundaries	39	12	6,597	508/733
Equal arcs shifted by +0.1 degree	39	12	6,597	508/733

The two J2000 partitions, H20 and H21, are specified on page 6. For the equal-grid shift, only the 27 mansion boundaries move. Section 2 proves the formula for C for every mansion partition satisfying §1. Section 3 computes D for the four rows above.

2. From a face to a distinct record

The coordinates (x,h) parameterize the torus bijectively. The intersection of a given mansion and rashi is empty or one arc: their total length is below one turn, so both can be lifted to intervals by cutting the circle outside their union. Any nonempty open intersection is a single arc of the combined partition. For a fixed tithi index t, the possible half-tithis j are 2t, 2t+1, 2t+30 and 2t+31. Their positive pairwise differences are 1, 29, 30 or 31. None is divisible by seven. Two movable karanas among them cannot share a name, and each of the four fixed names occurs only once. Thus (T,K) determines j.

Fixing N, R, K and T now gives one rectangle in lifts of (x,h). Inside it, s = 2x − h ranges over an interval of length at most 2/12 + 1/60 = 11/60. The gap between different lifts of the same yoga bin is 26/27. Only one lift of that bin can meet the rectangle. The intersection is convex, hence connected. Every label seen on a boundary also occurs in the interior: keep y fixed and increase x slightly. Each of x, h and s moves forward, preserving all half-open labels. The labels are constant on each open face. All interior witnesses of a tuple lie in a single connected strip intersection, so the open faces and realizable records correspond one-to-one.

There are 60v open rectangles before adding the yoga boundaries. Each of the 27 yoga circles h = 2x − i/27 crosses v vertical boundaries once and 60 horizontal boundaries twice. Each distinct crossing begins one open segment of the circle. In the open rectangles the resulting parallel line segments each add one face. A crossing at a corner was counted twice. At a fixed vertical boundary u the corner condition is

$$2u = i/27 + j/60 \pmod 1.$$

Set m = 1080u, with u in turns. A corner requires m to be an integer. For integral m, the homomorphism (i,j) ↦ 20i + 9j modulo 540 from residues modulo 27 and 60 is onto because 20 and 9 are coprime. Each m therefore has 27×60/540 = 3 preimages. The correction is three corners at each of q lattice boundaries. Hence C = 60v + 27(v + 120) − 3q.

In the unshifted equal case the shared mansion–rashi boundaries are 0°, 120° and 240°. All 36 distinct boundaries lie on the one-third-degree lattice. In each fixed table on page 6, the integer numerator of every mansion boundary (denominator 100,000,000 in degrees) ends in 44. Tripling it gives a numerator ending in 32, so no mansion boundary is an integer multiple of one third of a degree. Only the 12 rashi boundaries contribute to q. The +0.1° shift also moves every equal mansion boundary off the lattice.

3. Counting one-step errors

For each coordinate let E count its unordered pairs of valid records one step apart with all four other coordinates fixed. Adjacency here is measured in the cyclic name order. Write E_{total} for their sum. Each pair contributes two undetected directed edits. All five alphabets have more than two labels, so each realized record has ten distinct directed trials, and D = 1 − E_{total}/(5C).

Nakshatra and rashi. An adjacent pair of mansion bins can meet the interior of a common rashi only if its shared boundary b is inside that rashi. This follows by cutting the circle outside those three short arcs. The same statement holds with rashi and mansion exchanged. A shared mansion–rashi boundary contributes no pair changing just one name. For an eligible b, let the two adjoining x intervals be (a,b),(b,d), and fix one half-tithi interval (p,r). The attainable s intervals on the two sides overlap on (2b−r, 2b−p). A yoga bin occurs on both sides exactly when it meets this overlap: the intervals on each side overlap, and three pairwise intersecting intervals on a line have a common point. The span involved is at most 2/12 + 1/60, so different lifts of the same yoga bin cannot give an additional pair.

Over all 60 half-tithis these overlaps form the open bins of a translated 60-bin partition of the s circle. The number of two-sided yoga choices is 60 + 27 minus the number of coincident endpoints. The corner congruence above counts those coincidences. Thus each eligible b on the lattice contributes 84 pairs, and each eligible b off it contributes 87. The changed names identify b and (T,K) identifies j, so no pair is counted twice.

Yoga. Each of the 60v rectangles realizes a consecutive proper run of r yoga labels. The bound 11/60 from §2 ensures some yoga labels are absent. This gives r − 1 neighbouring pairs, including runs crossing index zero. Summing over rectangles gives $E_Y = C - 60v$.

Karana. At fixed tithi, half-tithis in opposite fortnights differ by 29, 30 or 31 slots. Their h values must therefore be at least 28/60 turn apart on the circle. But records with common rashi and yoga have circular h distance less than 2/12 + 1/27, by h = 2x − s. Only pairs j,j+1 with j even remain. Of the 30 candidates, j = 14, 28, 42 give K transitions 7 to 1, outside the specified one-step order. The remaining 27 qualify, at boundaries c = k/60 with odd k.

For such a boundary and fixed combined x arc (a,b), the yoga bins attained on both sides are those meeting (2a−c, 2b−c). The interval-intersection argument above applies, now with total span at most 2/12 + 2/60. Traversing the x circle, 2x−c encounters the 27 yoga boundaries twice each. Summing over all x arcs gives v + 54 minus the coincidences with vertical boundaries. A coincidence at u requires m = 1080u to be integral and to satisfy

$$m \equiv 20i + 9k \pmod{540}.$$

The right side is odd. For any rashi boundary u=n/12 or unshifted equal mansion boundary u=n/27, 1080u is even. The other mansion boundaries of the three comparison rows give nonintegral 1080u. Thus there is no coincidence in these cases, and $E_K = 27(v + 54)$. Given both records, (N,R) recovers the x arc and each (T,K) pair recovers its j, which identifies the boundary uniquely. A different partition can have a mansion boundary with odd 1080u, so this last count uses the stated inputs.

Tithi. If T differs by one while K stays fixed, the half-tithis must have the same movable karana, so their positive slot difference d is one of 7,14,21,28,35,42,49. Neighbouring tithi indices require d modulo 30 to belong to {1,2,3,27,28,29}. Only 28 qualifies. Such half-tithis are at least 27/60 turn apart, again exceeding the bound 2/12 + 1/27 from a common rashi and yoga. Hence $E_T = 0$; every tithi-only edit is detected.

Unordered pairs of realizable records one step apart

N partition	E_N	E_R	E_Y	E_K	E_T	E_total
Equal arcs, origin zero	2,016	756	4,104	2,430	0	9,306
Each other row above	2,349	1,008	4,257	2,511	0	10,125

The unshifted grid thus detects 44,028 of 62,640 directed edits (70.2874%). Each other partition detects 45,720 of 65,970 (69.3042%). Shifting equal-width mansions alone reproduces both counts for the J2000 inputs. Boundary placement therefore matters even when all widths are held fixed.

4. Relation to earlier work

Chamber counts for toric arrangements are established [2–4]. HV27 gave the equal-grid face count [1, §6] and explained why identifying faces with labelled records needs a separate proof [1, §7]. Here each face has its own five-label record, including with the specified unequal mansion arcs. We then count pairs of such records connected by one cyclic edit. Single-step neighbourhoods on a discrete torus occur in isoperimetric theory [5]. Our count retains the multiplicity of edits that start at different records and reach the same impossible record.

Jacobi verified the internal consistency of a specified historical date [6, §38]. Sewell and Dikshit described checks using other calendar elements [7, §§131–132]. Franceschini measured discrepancies in manuscript dates [8]. Our probabilities use a uniform distribution over realizable records and ten specified edits per record. Measuring historical error frequencies calls for an observed sample and its own error model.

Contribution and review status. Rushikesh Murlidhar Walke proposed the research direction and initial hypotheses. AI assistance, including ChatGPT, contributed to the proof arguments, pair counts, literature comparison, internal checks and writing. This preprint has not undergone independent human peer review.

References

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[8] M. Franceschini, On the Wrong Dates in the Colophons of Tamil Nadu Manuscripts: Data, Observations, Interpretative Hypotheses, in Praśasti (Universitätsverlag Halle-Wittenberg, 2025), 253–273, especially pp. 256, 259.

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A. Fixed mansion boundaries

The two columns give the right endpoint of each mansion in sidereal degrees at J2000.0 (1 January 2000, 12:00 TT). Read each column cyclically. Row 1 is Ashvini, row 27 is Revati. A row begins at the preceding row’s endpoint and ends just before its own right endpoint.

The displayed decimals define the two numerical examples. Each boundary is a circular midpoint of neighbouring stellar longitudes, with the adopted offset of 23.85305556° subtracted modulo 360°. In row 6 (Ardra), H20 uses α Ori and H21 uses γ Gem.

Row	H20 (°)	H21 (°)
1	17.23194444	17.23194444
2	30.24194444	30.24194444
3	41.03694444	41.03694444
4	52.89694444	52.89694444
5	62.37694444	67.55194444
6	77.13194444	82.30694444
7	97.11694444	97.11694444
8	106.67694444	106.67694444
9	117.23194444	117.23194444
10	131.72194444	131.72194444
11	142.61694444	142.61694444
12	158.68194444	158.68194444
13	174.79194444	174.79194444
14	180.18194444	180.18194444
15	190.80194444	190.80194444
16	209.97194444	209.97194444
17	222.31194444	222.31194444
18	233.32194444	233.32194444
19	245.73194444	245.73194444
20	254.62694444	254.62694444
21	268.22694444	268.22694444
22	285.20694444	285.20694444
23	305.10694444	305.10694444
24	323.68194444	323.68194444
25	337.47194444	337.47194444
26	350.66509744	350.66509744
27	3.07009744	3.07009744

How to read the boundary at zero

Revati begins at 350.66509744° and continues through 0° to 3.07009744°. At exactly 3.07009744° the label is Ashvini. Rows 5 and 6 supply the two right boundaries whose positions differ between H20 and H21.

The largest widths in the respective columns are 19.985° (H20, row 7) and 19.900° (H21, row 23). The reference equal grid has width 360°/27 and begins at zero. The fourth configuration shifts that entire mansion grid forward by 0.1° while keeping the other partitions fixed.